Chapter 5. Inference about a Mean Vector

Hotelling's T^2, likelihood ratio tests, confidence regions and simultaneous intervals, large-sample inference

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1 Starting Point: the Univariate t and Known $\boldsymbol{\Sigma}$

In one dimension $H_0 : \mu = \mu_0$ is tested with $t = \sqrt{n}(\bar{X} - \mu_0)/s$, and

$$t_{n-1}^2 = \sqrt{n}(\bar{X} - \mu_0)\,(s^2)^{-1}\sqrt{n}(\bar{X} - \mu_0) = \begin{pmatrix}\text{normal}\\ \text{r.v.}\end{pmatrix}\begin{pmatrix}\text{(scaled) chi-square}\\ \text{r.v./d.f.}\end{pmatrix}^{-1}\begin{pmatrix}\text{normal}\\ \text{r.v.}\end{pmatrix}.$$

If $\boldsymbol{\Sigma}$ is known, $\sqrt{n}\boldsymbol{\Sigma}^{-1/2}(\bar{\mathbf{X}}_n - \boldsymbol{\mu}) \sim N_p(\mathbf{0}, \mathbf{I}_p)$, so

$$n(\bar{\mathbf{X}}_n - \boldsymbol{\mu})^\top \boldsymbol{\Sigma}^{-1}(\bar{\mathbf{X}}_n - \boldsymbol{\mu}) \sim \chi_p^2, \qquad C_\alpha = \{\boldsymbol{\mu} \in \mathbb{R}^p : (\bar{\mathbf{X}}_n - \boldsymbol{\mu})^\top (\tfrac{1}{n}\boldsymbol{\Sigma})^{-1}(\bar{\mathbf{X}}_n - \boldsymbol{\mu}) \le \chi_{p,\alpha}^2\}.$$

Replacing $\boldsymbol{\Sigma}$ by $\mathbf{S}$ gives Hotelling's T^2.

2 Hotelling's T^2 Statistic

Motivation. Testing $\boldsymbol{\mu} = \boldsymbol{\mu}_0$ with p variables by p separate t-tests ignores correlation and inflates error. Hotelling's T^2 is the multivariate t-statistic: it combines all coordinates using $\mathbf{S}^{-1}$ and has an exact F distribution.

Definition 5.1: Hotelling's T^2

Suppose $\mathbf{X}$ and $\mathbf{S}$ are independent with $\mathbf{X} \sim N_p(\boldsymbol{\mu}, \boldsymbol{\Sigma})$ and $m\mathbf{S} \sim W_p(m, \boldsymbol{\Sigma})$. Then

$$T_p^2(m) = (\mathbf{X} - \boldsymbol{\mu})^\top \mathbf{S}^{-1}(\mathbf{X} - \boldsymbol{\mu})$$

is Hotelling's T^2 statistic. It plays the role of Student's t when $\boldsymbol{\Sigma}$ is *unknown*:

$$T_p^2(n-1) = \begin{pmatrix}\text{multivariate normal}\\ \text{random vector}\end{pmatrix}^\top \begin{pmatrix}\text{Wishart random}\\ \text{matrix/d.f.}\end{pmatrix}^{-1} \begin{pmatrix}\text{multivariate normal}\\ \text{random vector}\end{pmatrix}$$

$$= N_p(\mathbf{0}, \boldsymbol{\Sigma})^\top \left[\tfrac{1}{n-1} W_p(n-1, \boldsymbol{\Sigma})\right]^{-1} N_p(\mathbf{0}, \boldsymbol{\Sigma}).$$

Theorem 5.2: T^2 and the F distribution

If $m > p - 1$, then $\dfrac{m - p + 1}{mp} T_p^2(m) \sim F_{p, m-p+1}$.

Proof. Multiplying and dividing by $(\mathbf{X} - \boldsymbol{\mu})^\top \boldsymbol{\Sigma}^{-1}(\mathbf{X} - \boldsymbol{\mu})$,

$$\frac{m - p + 1}{mp} T_p^2(m) = \frac{(\mathbf{X} - \boldsymbol{\mu})^\top \boldsymbol{\Sigma}^{-1}(\mathbf{X} - \boldsymbol{\mu})/p}{\dfrac{(\mathbf{X} - \boldsymbol{\mu})^\top \boldsymbol{\Sigma}^{-1}(\mathbf{X} - \boldsymbol{\mu})}{(\mathbf{X} - \boldsymbol{\mu})^\top (m\mathbf{S})^{-1}(\mathbf{X} - \boldsymbol{\mu})} \Big/ (m - p + 1)} = \frac{Y/p}{W/(m - p + 1)},$$

where $Y = (\mathbf{X} - \boldsymbol{\mu})^\top \boldsymbol{\Sigma}^{-1}(\mathbf{X} - \boldsymbol{\mu}) \sim \chi_p^2$ (Example 3.22) and $W = \dfrac{\mathbf{a}^\top \boldsymbol{\Sigma}^{-1}\mathbf{a}}{\mathbf{a}^\top (m\mathbf{S})^{-1}\mathbf{a}}$ with the random vector $\mathbf{a} = \mathbf{X} - \boldsymbol{\mu}$ independent of $m\mathbf{S} \sim W_p(m, \boldsymbol{\Sigma})$; by Theorem 4.5(iii) $W \sim \chi_{m-p+1}^2$ and is independent of $\mathbf{a}$, hence of Y. By definition the ratio is $F_{p, m-p+1}$. ∎

Theorem 5.3: one-sample T^2

If $\mathbf{X}_1, \ldots, \mathbf{X}_n \overset{\text{iid}}{\sim} N_p(\boldsymbol{\mu}, \boldsymbol{\Sigma})$, $\bar{\mathbf{X}} = \frac{1}{n}\sum \mathbf{X}_j$, $\mathbf{S} = \frac{1}{n-1}\sum(\mathbf{X}_j - \bar{\mathbf{X}})(\mathbf{X}_j - \bar{\mathbf{X}})^\top$, then $\sqrt{n}(\bar{\mathbf{X}} - \boldsymbol{\mu}) \sim N_p(\mathbf{0}, \boldsymbol{\Sigma})$ and $(n-1)\mathbf{S} \sim W_p(n-1, \boldsymbol{\Sigma})$ independently, so

$$n(\bar{\mathbf{X}} - \boldsymbol{\mu})^\top \mathbf{S}^{-1}(\bar{\mathbf{X}} - \boldsymbol{\mu}) \sim T^2_p(n-1), \qquad n(\bar{\mathbf{X}} - \boldsymbol{\mu})^\top \mathbf{S}^{-1}(\bar{\mathbf{X}} - \boldsymbol{\mu}) \sim \frac{(n-1)p}{n-p} F_{p,n-p}.$$

($m = n-1$ gives $\frac{m-p+1}{mp} = \frac{n-p}{(n-1)p}$.) Hence $H_0 : \boldsymbol{\mu} = \boldsymbol{\mu}_0$ is rejected at level α if

$$T^2 = n(\bar{\mathbf{x}} - \boldsymbol{\mu}_0)^\top \mathbf{S}^{-1}(\bar{\mathbf{x}} - \boldsymbol{\mu}_0) > \frac{(n-1)p}{n-p} F_{p,n-p}(\alpha),$$

with p-value $P(F_{p,n-p} \geq \frac{n-p}{(n-1)p} T^2_{\text{obs}})$.

Theorem 5.4: invariance of T^2 under linear transformations

The T^2 statistic is unchanged by changes of units of the form $\mathbf{Y}_{(p\times 1)} = \mathbf{C}_{(p\times p)}\mathbf{X}_{(p\times 1)} + \mathbf{d}_{(p\times 1)}$ with $\mathbf{C}$ non-singular: $\bar{\mathbf{y}} = \mathbf{C}\bar{\mathbf{x}} + \mathbf{d}$, $\mathbf{S}_y = \mathbf{C}\mathbf{S}\mathbf{C}^\top$, $\boldsymbol{\mu}_Y = \mathbf{C}\boldsymbol{\mu}_X + \mathbf{d}$, and

$$n(\bar{\mathbf{y}} - \boldsymbol{\mu}_Y)^\top \mathbf{S}_y^{-1}(\bar{\mathbf{y}} - \boldsymbol{\mu}_Y) = n(\bar{\mathbf{x}} - \boldsymbol{\mu}_x)^\top \mathbf{S}^{-1}(\bar{\mathbf{x}} - \boldsymbol{\mu}_x) \sim T^2_p(n-1).$$

Proof. $n(\mathbf{C}(\bar{\mathbf{x}} - \boldsymbol{\mu}_x))^\top(\mathbf{C}\mathbf{S}\mathbf{C}^\top)^{-1}\mathbf{C}(\bar{\mathbf{x}} - \boldsymbol{\mu}_x) = n(\bar{\mathbf{x}} - \boldsymbol{\mu}_x)^\top \mathbf{C}^\top \mathbf{C}^{-\top}\mathbf{S}^{-1}\mathbf{C}^{-1}\mathbf{C}(\bar{\mathbf{x}} - \boldsymbol{\mu}_x)$. ■

Example 5.5: testing a mean vector with T^2 (sweat data)

Perspiration from $n = 20$ healthy females: X_1 sweat rate, X_2 sodium, X_3 potassium. Test $H_0 : \boldsymbol{\mu}^\top = [4, 50, 10]$ at $\alpha = 0.10$.

$$\bar{\mathbf{x}} = \begin{pmatrix} 4.640 \\ 45.400 \\ 9.965 \end{pmatrix}, \quad \mathbf{S} = \begin{pmatrix} 2.879 & 10.010 & -1.810 \\ 10.010 & 199.788 & -5.640 \\ -1.810 & -5.640 & 3.628 \end{pmatrix}, \quad \mathbf{S}^{-1} = \begin{pmatrix} .586 & -.022 & .258 \\ -.022 & .006 & -.002 \\ .258 & -.002 & .402 \end{pmatrix}.$$

$\bar{\mathbf{x}} - \boldsymbol{\mu}_0 = (.640, -4.600, -.035)^\top$ and $\mathbf{S}^{-1}(\bar{\mathbf{x}} - \boldsymbol{\mu}_0) = (.467, -.042, .160)^\top$, so $T^2 = 20(.640 \cdot .467 + 4.600 \cdot .042 - .035 \cdot .160) = 9.74$. Critical value $\frac{(n-1)p}{n-p} F_{p,n-p}(.10) = \frac{19(3)}{17} F_{3,17}(.10) = 3.353(2.44) = 8.18$. Since $9.74 > 8.18$, reject H_0 at the 10% level.

3 Likelihood Ratio Tests

3.1 $\boldsymbol{\Sigma}$ unknown

Under $H_0 : \boldsymbol{\mu} = \boldsymbol{\mu}_0$ the likelihood is

$$L(\boldsymbol{\mu}_0, \boldsymbol{\Sigma}) = \frac{1}{(2\pi)^{np/2}|\boldsymbol{\Sigma}|^{n/2}} \exp\Big(-\frac{1}{2}\sum_{j=1}^{n}(\mathbf{x}_j - \boldsymbol{\mu}_0)^\top \boldsymbol{\Sigma}^{-1}(\mathbf{x}_j - \boldsymbol{\mu}_0)\Big),$$

and the exponent equals $-\frac{1}{2}\text{tr}[\boldsymbol{\Sigma}^{-1}\sum_j(\mathbf{x}_j - \boldsymbol{\mu}_0)(\mathbf{x}_j - \boldsymbol{\mu}_0)^\top]$. Maximizing over $\boldsymbol{\Sigma}$ as in the proof of Theorem 4.7 (with $\sum_j(\mathbf{x}_j - \boldsymbol{\mu}_0)(\mathbf{x}_j - \boldsymbol{\mu}_0)^\top$ in place of $\sum_j(\mathbf{x}_j - \bar{\mathbf{x}})(\mathbf{x}_j - \bar{\mathbf{x}})^\top$),

$$\max_{\boldsymbol{\Sigma}} L(\boldsymbol{\mu}_0, \boldsymbol{\Sigma}) = \frac{1}{(2\pi)^{np/2}|\hat{\boldsymbol{\Sigma}}_0|^{n/2}} e^{-np/2}, \qquad \hat{\boldsymbol{\Sigma}}_0 = \frac{1}{n}\sum_{j=1}^{n}(\mathbf{x}_j - \boldsymbol{\mu}_0)(\mathbf{x}_j - \boldsymbol{\mu}_0)^\top.$$

The unrestricted maximum has the same form with $\hat{\boldsymbol{\Sigma}} = \frac{1}{n}\sum(\mathbf{x}_j - \bar{\mathbf{x}})(\mathbf{x}_j - \bar{\mathbf{x}})^\top$, so the *likelihood ratio statistic* is

$$\Lambda = \frac{\max_{\boldsymbol{\Sigma}} L(\boldsymbol{\mu}_0, \boldsymbol{\Sigma})}{\max_{\boldsymbol{\mu},\boldsymbol{\Sigma}} L(\boldsymbol{\mu}, \boldsymbol{\Sigma})} = \Big(\frac{|\hat{\boldsymbol{\Sigma}}|}{|\hat{\boldsymbol{\Sigma}}_0|}\Big)^{n/2}, \qquad \Lambda^{2/n} = \frac{|\hat{\boldsymbol{\Sigma}}|}{|\hat{\boldsymbol{\Sigma}}_0|} = \frac{|\sum_j (\mathbf{x}_j - \bar{\mathbf{x}})(\mathbf{x}_j - \bar{\mathbf{x}})^\top|}{|\sum_j (\mathbf{x}_j - \boldsymbol{\mu}_0)(\mathbf{x}_j - \boldsymbol{\mu}_0)^\top|}.$$

$\Lambda^{2/n}$ is *Wilks' lambda*. Reject H_0 if $\Lambda < c_\alpha$, the lower 100αth percentile of its distribution; fortunately the distribution of $\Lambda^{2/n}$ is available.

Theorem 5.6: likelihood ratio test and T^2

For a random sample from $N_p(\boldsymbol{\mu}, \boldsymbol{\Sigma})$, the T^2 test is equivalent to the likelihood ratio test of $H_0 : \boldsymbol{\mu} = \boldsymbol{\mu}_0$ versus $H_1 : \boldsymbol{\mu} \neq \boldsymbol{\mu}_0$ because

$$\Lambda^{2/n} = \Big(1 + \frac{T^2}{n-1}\Big)^{-1}.$$

H_0 is rejected for small $\Lambda^{2/n}$, equivalently large T^2.

First proof: block determinant. Let $\mathbf{Q} = \sum_j (\mathbf{x}_j - \bar{\mathbf{x}})(\mathbf{x}_j - \bar{\mathbf{x}})^\top = (n-1)\mathbf{S}$, $\boldsymbol{\delta} = \bar{\mathbf{x}} - \boldsymbol{\mu}_0$, and consider the $(p+1) \times (p+1)$ matrix

$$\mathbf{A} = \begin{pmatrix} \mathbf{Q} & \sqrt{n}\,\boldsymbol{\delta} \\ \sqrt{n}\,\boldsymbol{\delta}^\top & -1 \end{pmatrix} = \begin{pmatrix} \mathbf{A}_{11} & \mathbf{A}_{12} \\ \mathbf{A}_{21} & \mathbf{A}_{22} \end{pmatrix}.$$

From $|\mathbf{A}| = |\mathbf{A}_{22}||\mathbf{A}_{11} - \mathbf{A}_{12}\mathbf{A}_{22}^{-1}\mathbf{A}_{21}| = |\mathbf{A}_{11}||\mathbf{A}_{22} - \mathbf{A}_{21}\mathbf{A}_{11}^{-1}\mathbf{A}_{12}|$,

$$(-1)|\mathbf{Q} + n\boldsymbol{\delta}\boldsymbol{\delta}^\top| = |\mathbf{Q}|(-1 - n\boldsymbol{\delta}^\top\mathbf{Q}^{-1}\boldsymbol{\delta}) = -|\mathbf{Q}|\Big(1 + \frac{T^2}{n-1}\Big).$$

Also $\sum_j (\mathbf{x}_j - \boldsymbol{\mu}_0)(\mathbf{x}_j - \boldsymbol{\mu}_0)^\top = \sum_j (\mathbf{x}_j - \bar{\mathbf{x}} + \bar{\mathbf{x}} - \boldsymbol{\mu}_0)(\cdot)^\top = \mathbf{Q} + n\boldsymbol{\delta}\boldsymbol{\delta}^\top$. Hence $|n\hat{\boldsymbol{\Sigma}}_0| = |n\hat{\boldsymbol{\Sigma}}|(1 + \frac{T^2}{n-1})$ and $\Lambda^{2/n} = |\hat{\boldsymbol{\Sigma}}|/|\hat{\boldsymbol{\Sigma}}_0| = (1 + \frac{T^2}{n-1})^{-1}$. ■

Second proof: determinant lemma. With $\boldsymbol{\delta} = \bar{\mathbf{x}} - \boldsymbol{\mu}_0$ and $\mathbf{S}_0 = \hat{\boldsymbol{\Sigma}}$, $\hat{\boldsymbol{\Sigma}}_0 = \mathbf{S}_0 + \boldsymbol{\delta}\boldsymbol{\delta}^\top$ and $|\mathbf{S}_0 + \boldsymbol{\delta}\boldsymbol{\delta}^\top| = |\mathbf{S}_0^{1/2}||\mathbf{I}_p + \mathbf{S}_0^{-1/2}\boldsymbol{\delta}\boldsymbol{\delta}^\top\mathbf{S}_0^{-1/2}||\mathbf{S}_0^{1/2}| = |\mathbf{S}_0|(1 + \boldsymbol{\delta}^\top\mathbf{S}_0^{-1}\boldsymbol{\delta})$ using $|\mathbf{I} + \mathbf{u}\mathbf{v}^\top| = 1 + \mathbf{v}^\top\mathbf{u}$. Since $\mathbf{S}_0 = \frac{n-1}{n}\mathbf{S}$, $\boldsymbol{\delta}^\top\mathbf{S}_0^{-1}\boldsymbol{\delta} = \frac{n}{n-1}\boldsymbol{\delta}^\top\mathbf{S}^{-1}\boldsymbol{\delta} = \frac{T^2}{n-1} =: \frac{T_0^2}{n-1}$. ■

Remark 5.7: approximation of $W = -2\log\Lambda$

$W = -2\log(L_0/L_1) = n\log(1 + \frac{T^2}{n-1})$. Under H_0, $T^2 = O_p(1)$, so $W \approx \frac{n}{n-1}T^2 \approx T^2$, and for large n, $\frac{np}{n-p}F_{p,n-p} \approx \chi^2_p$: $W \dot{\sim} \chi^2_p$, in agreement with general likelihood ratio theory. Exact test: reject if $T_0^2 > \frac{(n-1)p}{n-p}F_{p,n-p,\alpha}$; approximate test: reject if $W > \chi^2_{p,\alpha}$.

3.2 $\boldsymbol{\Sigma}$ known

With $L_0 = L(\boldsymbol{\mu}_0, \boldsymbol{\Sigma})$ and $L_1 = L(\bar{\mathbf{x}}, \boldsymbol{\Sigma})$, $W = -2\log L_0 - (-2\log L_1) = n(\bar{\mathbf{x}} - \boldsymbol{\mu}_0)^\top\boldsymbol{\Sigma}^{-1}(\bar{\mathbf{x}} - \boldsymbol{\mu}_0)$, which is exactly χ^2_p under H_0.

3.3 Testing a covariance matrix: $H_0 : \boldsymbol{\Sigma} = \boldsymbol{\Sigma}_0$, $\boldsymbol{\mu}$ unknown

Theorem 5.8: likelihood ratio test for $\boldsymbol{\Sigma} = \boldsymbol{\Sigma}_0$

Let $\mathbf{X}_1, \ldots, \mathbf{X}_n \overset{\text{iid}}{\sim} N_p(\boldsymbol{\mu}, \boldsymbol{\Sigma})$ with $\boldsymbol{\mu}$ unknown, and $\boldsymbol{\Sigma}_0 \succ 0$ given. For $H_0 : \boldsymbol{\Sigma} = \boldsymbol{\Sigma}_0$ versus $H_1 : \boldsymbol{\Sigma} \neq \boldsymbol{\Sigma}_0$,

$$W = -2 \log \Lambda = -n \log |\boldsymbol{\Sigma}_0^{-1} \mathbf{S}_0| - np + n \operatorname{tr}(\boldsymbol{\Sigma}_0^{-1} \mathbf{S}_0) \;\dot{\sim}\; \chi^2_\nu, \qquad \nu = \frac{p(p+1)}{2},$$

where $\mathbf{S}_0 = \frac{1}{n} \sum_i (\mathbf{x}_i - \bar{\mathbf{x}})(\mathbf{x}_i - \bar{\mathbf{x}})^\top$. Equivalently, with $l_1, \ldots, l_p$ the eigenvalues of $\boldsymbol{\Sigma}_0^{-1} \mathbf{S}_0$,

$$W = n \sum_{i=1}^{p} (l_i - \log l_i - 1) \;\geq 0.$$

Reject H_0 if $W > \chi^2_\nu(\alpha)$.

Proof [Supplement]. Step 1 (the statistic). By Theorem 4.7, $-2 \log L(\boldsymbol{\mu}, \boldsymbol{\Sigma}) = n \log |\boldsymbol{\Sigma}| + n(\bar{\mathbf{x}} - \boldsymbol{\mu})^\top \boldsymbol{\Sigma}^{-1} (\bar{\mathbf{x}} - \boldsymbol{\mu}) + n \operatorname{tr}(\boldsymbol{\Sigma}^{-1} \mathbf{S}_0) + c$. Under H_1 the maximum is at $(\bar{\mathbf{x}}, \mathbf{S}_0)$: $-2 \log L_1 = n \log |\mathbf{S}_0| + np + c$. Under H_0 only $\boldsymbol{\mu}$ is free and its maximizer is $\bar{\mathbf{x}}$ whatever $\boldsymbol{\Sigma}_0$ is: $-2 \log L_0 = n \log |\boldsymbol{\Sigma}_0| + n \operatorname{tr}(\boldsymbol{\Sigma}_0^{-1} \mathbf{S}_0) + c$. Hence

$$\begin{aligned} W = -2 \log L_0 + 2 \log L_1 &= n \log |\boldsymbol{\Sigma}_0| - n \log |\mathbf{S}_0| + n \operatorname{tr}(\boldsymbol{\Sigma}_0^{-1} \mathbf{S}_0) - np \\ &= -n \log |\boldsymbol{\Sigma}_0^{-1} \mathbf{S}_0| - np + n \operatorname{tr}(\boldsymbol{\Sigma}_0^{-1} \mathbf{S}_0). \end{aligned}$$

The eigenvalues l_i of $\boldsymbol{\Sigma}_0^{-1} \mathbf{S}_0$ equal those of the PD matrix $\boldsymbol{\Sigma}_0^{-1/2} \mathbf{S}_0 \boldsymbol{\Sigma}_0^{-1/2}$, so $l_i > 0$, $|\boldsymbol{\Sigma}_0^{-1} \mathbf{S}_0| = \prod l_i$, $\operatorname{tr}(\boldsymbol{\Sigma}_0^{-1} \mathbf{S}_0) = \sum l_i$, giving $W = n \sum (l_i - \log l_i - 1) \geq 0$ by Theorem 2.13, with $W = 0$ iff $\mathbf{S}_0 = \boldsymbol{\Sigma}_0$.

Step 2 (reduction to $\boldsymbol{\Sigma}_0 = \mathbf{I}$). Put $\mathbf{Y}_i = \boldsymbol{\Sigma}_0^{-1/2} \mathbf{X}_i$. Then $\mathbf{S}_0^{(Y)} = \boldsymbol{\Sigma}_0^{-1/2} \mathbf{S}_0 \boldsymbol{\Sigma}_0^{-1/2}$ has the same eigenvalues l_i, so W is unchanged, and under H_0, $\mathbf{Y}_i \overset{\text{iid}}{\sim} N_p(\boldsymbol{\Sigma}_0^{-1/2} \boldsymbol{\mu}, \mathbf{I}_p)$. So assume $\boldsymbol{\Sigma}_0 = \mathbf{I}$ and write $\mathbf{S}_0$ for the MLE.

Step 3 (quadratic approximation). Under H_0, $\mathbf{S}_0 \xrightarrow{p} \mathbf{I}$, so all $l_i \xrightarrow{p} 1$. Taylor expansion of $g(l) = l - \log l - 1$ at $l = 1$: $g(1) = g'(1) = 0$, $g''(1) = 1$, so $g(l) = \frac{1}{2}(l-1)^2 + O(|l-1|^3)$ and

$$W = \frac{n}{2} \sum_i (l_i - 1)^2 + n \, O_p\Big(\max_i |l_i - 1|^3 \Big) = \frac{n}{2} \|\mathbf{S}_0 - \mathbf{I}\|_F^2 + o_p(1),$$

using $\sum_i (l_i - 1)^2 = \operatorname{tr}\{(\mathbf{S}_0 - \mathbf{I})^2\} = \|\mathbf{S}_0 - \mathbf{I}\|_F^2$ and $\sqrt{n}(\mathbf{S}_0 - \mathbf{I}) = O_p(1)$ (Step 4), so that $n \max |l_i - 1|^3 = O_p(n^{-1/2})$.

Step 4 (limit of $\sqrt{n}(\mathbf{S}_0 - \mathbf{I})$). Write $\mathbf{S}_0 = \frac{1}{n} \sum_i \mathbf{Z}_i \mathbf{Z}_i^\top - (\bar{\mathbf{Z}})(\bar{\mathbf{Z}})^\top$ with $\mathbf{Z}_i = \mathbf{Y}_i - \mathrm{E}\mathbf{Y}_i \overset{\text{iid}}{\sim} N_p(\mathbf{0}, \mathbf{I})$; the second term is $O_p(1/n)$. For the vector of distinct entries $\{Z_j Z_k\}_{j \leq k}$, $\mathrm{E}(Z_j Z_k) = \delta_{jk}$ and

$$\operatorname{Cov}(Z_j Z_k, Z_l Z_m) = \mathrm{E}(Z_j Z_k Z_l Z_m) - \delta_{jk} \delta_{lm} = \delta_{jl} \delta_{km} + \delta_{jm} \delta_{kl},$$

so $\operatorname{Var}(Z_j^2) = 2$, $\operatorname{Var}(Z_j Z_k) = 1$ $(j < k)$, and different pairs $\{j, k\} \neq \{l, m\}$ are uncorrelated (consistent with Theorem 4.2(iv): $\Sigma_{jk}^2 + \Sigma_{jj} \Sigma_{kk}$). By the multivariate CLT,

$$\sqrt{n}(s_{jj} - 1) \Rightarrow N(0, 2), \qquad \sqrt{n}\, s_{jk} \Rightarrow N(0, 1) \;\; (j < k),$$

jointly, with a diagonal limiting covariance, i.e. all $p(p+1)/2$ limits are independent.

Step 5 (chi-square limit). By the CMT,

$$\frac{n}{2} \|\mathbf{S}_0 - \mathbf{I}\|_F^2 = \sum_{j=1}^{p} \frac{\{\sqrt{n}(s_{jj} - 1)\}^2}{2} + \sum_{j<k} \{\sqrt{n}\, s_{jk}\}^2 \Rightarrow \sum_{j=1}^{p} \frac{(\sqrt{2} Z_{jj})^2}{2} + \sum_{j<k} Z_{jk}^2 = \chi^2_{p + p(p-1)/2} = \chi^2_{p(p+1)/2},$$

with independent $Z_{jk} \sim N(0,1)$. By Slutsky, $W \Rightarrow \chi^2_{p(p+1)/2}$. The degrees of freedom $p(p+1)/2$ are the number of free parameters in a symmetric $\boldsymbol{\Sigma}$ fixed by H_0, as general likelihood ratio theory predicts. ■

Remark 5.9: special cases

- $\boldsymbol{\Sigma}_0 = \sigma_0^2 \mathbf{I}$: $W = n\{\mathrm{tr}(\mathbf{S}_0)/\sigma_0^2 - \log|\mathbf{S}_0/\sigma_0^2| - p\}$.
- $p = 1$: $W = n\{s_0^2/\sigma_0^2 - \log(s_0^2/\sigma_0^2) - 1\}$, an asymptotic version of the χ^2 test for a variance; the exact test uses $ns_0^2/\sigma_0^2 \sim \chi^2_{n-1}$.
- Bartlett-type corrections replace n by $n-1$ (using $\mathbf{S}$ instead of $\mathbf{S}_0$) and an additional factor to improve the χ^2 approximation.

3.4 Summary: likelihood ratio tests under normality

Hypothesis	Statistic and distribution
(i) $H_0 : \boldsymbol{\mu} = \boldsymbol{\mu}_0$, $\boldsymbol{\Sigma}$ known	$W = n(\bar{\mathbf{x}} - \boldsymbol{\mu}_0)^\top \boldsymbol{\Sigma}^{-1}(\bar{\mathbf{x}} - \boldsymbol{\mu}_0) \sim \chi^2_p$ (exact)
(ii) $H_0 : \boldsymbol{\mu} = \boldsymbol{\mu}_0$, $\boldsymbol{\Sigma}$ unknown	$W = n\log(1 + T^2/(n-1))$; exact: $T^2 \sim \frac{(n-1)p}{n-p} F_{p,n-p}$
(iii) $H_0 : \boldsymbol{\Sigma} = \boldsymbol{\Sigma}_0$, $\boldsymbol{\mu}$ unknown	$W = -n\log\lvert\boldsymbol{\Sigma}_0^{-1}\mathbf{S}_0\rvert - np + n\mathrm{tr}(\boldsymbol{\Sigma}_0^{-1}\mathbf{S}_0) \dot{\sim} \chi^2_{p(p+1)/2}$
(iv) $H_0 : \boldsymbol{\mu}_1 = \boldsymbol{\mu}_2$, common $\boldsymbol{\Sigma}$ unknown	$W = n\log\left(1 + \frac{T^2}{n_1+n_2-2}\right)$; exact: $\frac{n_1+n_2-p-1}{(n_1+n_2-2)p} T^2 \sim F_{p,n_1+n_2-p-1}$

Theorem 5.10: two-sample likelihood ratio test

Let $\mathbf{X}_{11}, \ldots, \mathbf{X}_{1n_1} \overset{\text{iid}}{\sim} N_p(\boldsymbol{\mu}_1, \boldsymbol{\Sigma})$ and $\mathbf{X}_{21}, \ldots, \mathbf{X}_{2n_2} \overset{\text{iid}}{\sim} N_p(\boldsymbol{\mu}_2, \boldsymbol{\Sigma})$ be independent, $n = n_1 + n_2$. For $H_0 : \boldsymbol{\mu}_1 = \boldsymbol{\mu}_2$ versus $H_1 : \boldsymbol{\mu}_1 \neq \boldsymbol{\mu}_2$ with $\boldsymbol{\Sigma}$ unknown,

$$\bar{\mathbf{X}}_1 - \bar{\mathbf{X}}_2 \sim N_p\Big(\boldsymbol{\mu}_1 - \boldsymbol{\mu}_2, \frac{n_1+n_2}{n_1 n_2}\boldsymbol{\Sigma}\Big), \qquad (n_1+n_2-2)\mathbf{S}_p \sim W_p(n_1+n_2-2, \boldsymbol{\Sigma}),$$

$$T^2(n_1+n_2-2) = \frac{n_1 n_2}{n_1+n_2}(\bar{\mathbf{X}}_1 - \bar{\mathbf{X}}_2)^\top \mathbf{S}_p^{-1}(\bar{\mathbf{X}}_1 - \bar{\mathbf{X}}_2),$$

$$\frac{n_1+n_2-p-1}{(n_1+n_2-2)p} T^2(n_1+n_2-2) \sim F_{p,n_1+n_2-p-1},$$

and the likelihood ratio satisfies $\Lambda^{2/n} = \left(1 + \frac{T^2}{n_1+n_2-2}\right)^{-1}$, so the LRT is the T^2 test.

Proof [Supplement]. The distributional facts are Theorem 6.5. Under H_1 the MLEs are $\bar{\mathbf{x}}_1, \bar{\mathbf{x}}_2$ and $\hat{\boldsymbol{\Sigma}}_1 = \mathbf{W}/n$, $\mathbf{W} = (n_1+n_2-2)\mathbf{S}_p$; under H_0 they are the pooled mean $\bar{\mathbf{x}}$ and $\hat{\boldsymbol{\Sigma}}_0 = (\mathbf{W}+\mathbf{B})/n$ with $\mathbf{B} = \frac{n_1 n_2}{n}\mathbf{d}\mathbf{d}^\top$, $\mathbf{d} = \bar{\mathbf{x}}_1 - \bar{\mathbf{x}}_2$ (the $g = 2$ case of Ch. 6). As in Theorem 5.6, $\Lambda^{2/n} = |\hat{\boldsymbol{\Sigma}}_1|/|\hat{\boldsymbol{\Sigma}}_0| = |\mathbf{W}|/|\mathbf{W}+\mathbf{B}|$, and the determinant lemma gives $|\mathbf{W}+\mathbf{B}| = |\mathbf{W}|(1 + \frac{n_1 n_2}{n}\mathbf{d}^\top \mathbf{W}^{-1}\mathbf{d}) = |\mathbf{W}|(1 + \frac{T^2}{n_1+n_2-2})$. ■

Note. The F constant is $\frac{n_1+n_2-p-1}{(n_1+n_2-2)p}$, from Theorem 5.2 with $m = n_1 + n_2 - 2$.

4 Confidence Regions

Motivation. Inverting the T^2 test gives a confidence ellipsoid for $\boldsymbol{\mu}$, whose axes are the eigenvectors of $\mathbf{S}$.

Theorem 5.11: $100(1-\alpha)\%$ confidence region for $\boldsymbol{\mu}$

$$R(\mathbf{X}) = \Big\{\boldsymbol{\mu} \in \mathbb{R}^p : \ \frac{n}{n-1}\frac{n-p}{p}(\bar{\mathbf{x}}-\boldsymbol{\mu})^\top \mathbf{S}_0^{-1}(\bar{\mathbf{x}}-\boldsymbol{\mu}) \le F_{p,n-p,\alpha}\Big\}$$
$$= \Big\{\boldsymbol{\mu} : n(\bar{\mathbf{x}}-\boldsymbol{\mu})^\top \mathbf{S}^{-1}(\bar{\mathbf{x}}-\boldsymbol{\mu}) \le \frac{(n-1)p}{n-p}F_{p,n-p}(\alpha)\Big\}.$$

With $\mathbf{S} = \sum \lambda_i \mathbf{e}_i \mathbf{e}_i^\top$ it is an ellipsoid centered at $\bar{\mathbf{x}}$ with axes along $\mathbf{e}_i$ and half-lengths $\sqrt{\lambda_i}\sqrt{\frac{(n-1)p}{n(n-p)}F_{p,n-p}(\alpha)}$. Checking whether $\boldsymbol{\mu}_0 \in R$ is the T^2 test.

For $p \ge 3$ the ellipsoid cannot be drawn, so componentwise intervals are needed.

5 Simultaneous Confidence Intervals

5.1 A single linear combination

For fixed $\mathbf{a}$, $\mathbf{a}^\top(\bar{\mathbf{X}}-\boldsymbol{\mu}) \sim N_1(0, \frac{\mathbf{a}^\top\boldsymbol{\Sigma}\mathbf{a}}{n})$ and $(n-1)\mathbf{a}^\top\mathbf{S}\mathbf{a} \sim W_1(n-1, \mathbf{a}^\top\boldsymbol{\Sigma}\mathbf{a}) \iff (n-1)\frac{\mathbf{a}^\top\mathbf{S}\mathbf{a}}{\mathbf{a}^\top\boldsymbol{\Sigma}\mathbf{a}} \sim \chi^2_{n-1}$, independent, so

$$t(\mathbf{a}) = \frac{\sqrt{n}\,\mathbf{a}^\top(\bar{\mathbf{X}}-\boldsymbol{\mu})/\sqrt{\mathbf{a}^\top\boldsymbol{\Sigma}\mathbf{a}}}{\sqrt{(n-1)\frac{\mathbf{a}^\top\mathbf{S}\mathbf{a}}{\mathbf{a}^\top\boldsymbol{\Sigma}\mathbf{a}}/(n-1)}} = \frac{\mathbf{a}^\top(\bar{\mathbf{X}}-\boldsymbol{\mu})}{\sqrt{\mathbf{a}^\top\mathbf{S}\mathbf{a}/n}} \sim t_{n-1}, \qquad C_\alpha = \Big\{\theta : \frac{|\mathbf{a}^\top\bar{\mathbf{x}}-\theta|}{\sqrt{\mathbf{a}^\top\mathbf{S}\mathbf{a}/n}} \le t_{n-1,\alpha/2}\Big\}.$$

Using these intervals for several $\mathbf{a}$ at once lowers the overall confidence below $1-\alpha$.

5.2 T^2 simultaneous intervals

Theorem 5.12: simultaneous confidence intervals for $\mathbf{a}^\top\boldsymbol{\mu}$

Let $\mathbf{X}_1, \dots, \mathbf{X}_n$ be a random sample from $N_p(\boldsymbol{\mu}, \boldsymbol{\Sigma})$ with $\boldsymbol{\Sigma} \succ 0$. Then, *simultaneously for all* $\mathbf{a}$, the interval

$$\Big(\mathbf{a}^\top\bar{\mathbf{X}} - \sqrt{\frac{p(n-1)}{n(n-p)}F_{p,n-p}(\alpha)\mathbf{a}^\top\mathbf{S}\mathbf{a}}, \ \ \mathbf{a}^\top\bar{\mathbf{X}} + \sqrt{\frac{p(n-1)}{n(n-p)}F_{p,n-p}(\alpha)\mathbf{a}^\top\mathbf{S}\mathbf{a}}\Big)$$

contains $\mathbf{a}^\top\boldsymbol{\mu}$ with probability $1-\alpha$.

Proof. Simple algebra gives, for every $\mathbf{a}$,

$$\mathbf{a}^\top\bar{\mathbf{x}} - c\sqrt{\frac{\mathbf{a}^\top\mathbf{S}\mathbf{a}}{n}} \le \mathbf{a}^\top\boldsymbol{\mu} \le \mathbf{a}^\top\bar{\mathbf{x}} + c\sqrt{\frac{\mathbf{a}^\top\mathbf{S}\mathbf{a}}{n}} \iff t(\mathbf{a})^2 = \frac{n(\mathbf{a}^\top\bar{\mathbf{x}}-\mathbf{a}^\top\boldsymbol{\mu})^2}{\mathbf{a}^\top\mathbf{S}\mathbf{a}} \le c^2.$$

The key is $\max_{\mathbf{a}} t(\mathbf{a})^2 = T^2$: with $\mathbf{b} = \mathbf{S}^{1/2}\mathbf{a}$ and $\mathbf{v} = \sqrt{n}\mathbf{S}^{-1/2}(\bar{\mathbf{x}}-\boldsymbol{\mu})$,

$$\max_{\mathbf{a}} t(\mathbf{a})^2 = \max_{\mathbf{b}} \frac{(\mathbf{b}^\top\mathbf{v})^2}{\mathbf{b}^\top\mathbf{b}} \le \mathbf{v}^\top\mathbf{v} = n(\bar{\mathbf{x}}-\boldsymbol{\mu})^\top\mathbf{S}^{-1}(\bar{\mathbf{x}}-\boldsymbol{\mu}) = T^2(n-1)$$

by Cauchy–Schwarz, with equality at $\mathbf{a} \propto \mathbf{S}^{-1}(\bar{\mathbf{x}}-\boldsymbol{\mu})$. By Theorem 5.2, choosing $c^2 = \frac{p(n-1)}{n-p}F_{p,n-p}(\alpha)$ gives $P(\bigcap_{\mathbf{a}}\{t(\mathbf{a})^2 \le c^2\}) = P(\max_{\mathbf{a}} t(\mathbf{a})^2 \le c^2) = P(T^2 \le c^2) = 1-\alpha$. ∎

- These are called T^2-intervals, since the coverage is determined by the distribution of T^2. With $\mathbf{a} = \mathbf{e}_j$: $\bar{x}_j \pm \sqrt{\frac{p(n-1)}{n-p}F_{p,n-p}(\alpha)}\sqrt{\frac{s_{jj}}{n}}$, holding simultaneously with confidence $1-\alpha$.

- With $\mathbf{a}^\top = [0,\dots,0,a_j,0,\dots,0,a_k,0,\dots]$, $a_j = 1$, $a_k = -1$: $\mathbf{a}^\top\mathbf{S}\mathbf{a} = s_{jj} - 2s_{jk} + s_{kk}$, giving intervals for $\mu_j - \mu_k$.
- The coefficient $1-\alpha$ holds for any choice of $\mathbf{a}$, so these intervals are ideal for "data snooping".
- The rectangle formed by the componentwise T^2 intervals is the shadow of the confidence ellipsoid on the coordinate axes.

Example 5.13: simultaneous intervals (microwave radiation data)

$n = 42$, $p = 2$, $\bar{\mathbf{x}} = (0.564, 0.603)^\top$, $s_{11} = 0.0144$, $s_{22} = 0.0146$, $F_{2,40}(.05) = 3.23$. The 95% simultaneous T^2 intervals are

$$0.564 \pm \sqrt{\tfrac{2(41)}{40}3.23}\sqrt{\tfrac{0.0144}{42}} = (0.516, 0.612), \qquad 0.603 \pm \sqrt{\tfrac{2(41)}{40}3.23}\sqrt{\tfrac{0.0146}{42}} = (0.555, 0.651).$$

5.3 The Bonferroni method

When attention is restricted to a small number m of pre-specified linear combinations, shorter (more precise) simultaneous intervals are possible.

Theorem 5.14: Bonferroni inequality

Let C_i be a confidence statement about $\mathbf{a}_i^\top\boldsymbol{\mu}$ with $P(C_i \text{ true}) = 1 - \alpha_i$. Then

$$P[\text{all } C_i \text{ true}] = 1 - P[\text{at least one } C_i \text{ false}] \geq 1 - \sum_{i=1}^{m} P(C_i \text{ false}) = 1 - (\alpha_1 + \cdots + \alpha_m).$$

With $\alpha_i = \alpha/m$, the individual intervals $\bar{x}_i \pm t_{n-1}(\frac{\alpha}{2m})\sqrt{s_{ii}/n}$, $i = 1,\dots,m$, hold simultaneously with confidence at least $1-\alpha$. (Equivalently, with $C_i(\alpha) = \{\mu_i : |\bar{x}_i - \mu_i|/(s_i/\sqrt{n}) < t_{n-1,\alpha/2}\}$; setting $\mu_i \in C_i(\alpha/m)$ gives $P(\mu_i \in C_i(\alpha/m),\ i = 1,\dots,m) \geq 1 - \sum_i P(\mu_i \notin C_i(\alpha/m)) = 1 - \alpha$.)

Remark 5.15: comparison of Bonferroni and T^2 interval lengths

Both have the form $\mathbf{a}^\top\bar{\mathbf{x}} \pm (\text{critical value})\sqrt{\mathbf{a}^\top\mathbf{S}\mathbf{a}/n}$, so with $\alpha_i = \alpha/m$

$$\frac{\text{length of Bonferroni interval}}{\text{length of } T^2 \text{ interval}} = \frac{t_{n-1}(\alpha/2m)}{\sqrt{\frac{p(n-1)}{n-p}F_{p,n-p}(\alpha)}},$$

which does not depend on $\bar{\mathbf{X}}$ or $\mathbf{S}$. For $1-\alpha = .95$, $\alpha_i = .05/m$, $m = p$:

n	$m=2$	$m=4$	$m=10$
15	.88	.69	.29
25	.90	.75	.48
50	.91	.78	.58
100	.91	.80	.62
∞	.91	.81	.66

For a small number of specified functions $\mathbf{a}^\top\boldsymbol{\mu}$ the Bonferroni intervals are shorter; being easy to apply, they are used often. In Example 5.13 the Bonferroni intervals are $(0.521, 0.607)$ and $(0.560, 0.646)$, inside the T^2 intervals.

6 Large-Sample Inference about a Mean Vector

Motivation. When normality is doubtful but n is large, T^2 is approximately χ^2_p, and the same procedures apply with χ^2 critical values.

Theorem 5.16: hypothesis testing for $\boldsymbol{\mu}$ with large $n-p$

Let $\mathbf{X}_1, \ldots, \mathbf{X}_n$ be a random sample from a population with mean $\boldsymbol{\mu}$ and positive definite covariance $\boldsymbol{\Sigma}$ (not necessarily normal). When $n-p$ is large, $H_0 : \boldsymbol{\mu} = \boldsymbol{\mu}_0$ is rejected in favor of $H_1 : \boldsymbol{\mu} \neq \boldsymbol{\mu}_0$, at level approximately α, if

$$n(\bar{\mathbf{x}} - \boldsymbol{\mu}_0)^\top \mathbf{S}^{-1}(\bar{\mathbf{x}} - \boldsymbol{\mu}_0) > \chi^2_p(\alpha).$$

The basis is $n(\bar{\mathbf{X}} - \boldsymbol{\mu})^\top \mathbf{S}^{-1}(\bar{\mathbf{X}} - \boldsymbol{\mu}) \Rightarrow \chi^2_p$ (Ch. 4). The statistic is the same as in normal theory; only the critical values differ, since for n large relative to p, $\frac{(n-1)pF_{p,n-p}(\alpha)}{n-p} \approx \chi^2_p(\alpha)$.

Theorem 5.17: large-sample simultaneous intervals

If $n-p$ is large, $\mathbf{a}^\top \bar{\mathbf{X}} \pm \sqrt{\chi^2_p(\alpha)}\sqrt{\mathbf{a}^\top \mathbf{S}\mathbf{a}/n}$ contains $\mathbf{a}^\top \boldsymbol{\mu}$ for every $\mathbf{a}$ with probability approximately $1-\alpha$. In particular $\bar{x}_j \pm \sqrt{\chi^2_p(\alpha)}\sqrt{s_{jj}/n}$ contains μ_j, $j = 1, \ldots, p$.

What counts as "large" is not easy to answer. Example: from the Golub data take two genes ($p = 2$) and the 11 AML and 27 ALL patients; the 95% and 99% elliptical regions and the simultaneous intervals (with $\mathbf{a}_1 = (1,0)^\top$, $\mathbf{a}_2 = (0,1)^\top$) show clearly different means for the two groups.

Further topic: multivariate quality control charts — the T^2 chart compares $T^2_j = (\mathbf{x}_j - \bar{\mathbf{x}})^\top \mathbf{S}^{-1}(\mathbf{x}_j - \bar{\mathbf{x}})$ with the control limit $\chi^2_p(\alpha)$.